MIL-D-10131B(EA)
18 October 1974
SUPERSEDING
MIL-D-10131A(MU)
24 July 1967

MILITARY SPECIFICATION

DISKS, INLET VALVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and two sizes of natural rubber valve disks.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

B5-2-535 - Mounting, Canister.
A5-2-537 - Disk, Inlet Valve.
A5-3-95 - Disk, Inlet Valve.
A5-3-450 - Seat, Inlet Valve.

FSC 4240

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DL136-14-170-4 - Tester, Air Leakage, Flow Type, Q211.
DL136-14-170-5 - Tester, Air Leakage, Flow Type, Q211.
DL136-44-1-5 - Indicator, Inlet and Outlet Valve Resistance, Q212.
DL136-44-1-6 - Indicator, Inlet and Outlet Valve Resistance, Q212.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM STANDARDS

D572-67 - Accelerated Aging of Vulcanized Rubber by the Oxygen-Pressure Method.

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber compound. The rubber compound used in the manufacture of the sheet stock shall contain not less than 92 percent, by volume, of new natural rubber.

3.1.2 Antioxidants and accelerators. The supplier shall furnish a certificate stating the trade names, the manufacturers, and the amount of the antioxidants and the accelerators used in compounding the stock.

3.2 Manufacture. The disks shall be manufactured from the molded sheet stock in accordance with the applicable drawing listed in 2.1.

3.2.1 Color. The color of the disks shall be as specified in the contract or order. (See 6.2.)

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3.3 Flexibility. When assembled to the appropriate valve seat the disks shall conform to the entire valve seat surface and shall not fall away at any point from the seat regardless of the position in which the seat is held.

3.4 Physical properties. Physical properties of the sheet stock shall conform to the requirements of table I when tested as specified therein.

Table I. Physical properties

Property	Unaged	After heat aging	Test paragraph
Average tensile strength, psi, min	2600		4.2.4.2
Percent increase, max		20	
Percent decrease, max		25	
Average ultimate elongation, %, min	650		4.2.4.2
Percent increase, max		20	
Percent decrease, max		25	
Average tension set at 600% elongation, % max	15		4.2.4.2

3.5 Leakage. When assembled to the appropriate valve seat the disks shall not leak more than 250 milliliters of air per minute at a pressure differential of 25 millimeters of water when tested as specified in 4.2.4.3.

3.6 Airflow resistance. When assembled to the appropriate valve seat the disks shall have a resistance not to exceed a pressure differential of 5 millimeters of water for the A5-3-95 disk and 6 millimeters of water for the A5-2-537 disk at an airflow rate of 85 liters per minute when tested as specified in 4.2.4.4.

3.7 Workmanship. The disks shall have a smooth finish on both sides; shall have clean cut peripheral and hole edges; and shall be free from tackiness, imperfections, such as porous areas, cracks, checks, abrasions and blisters, foreign matter (grease, oil, dirt, or ash), and damage, such as tears, slits, and punctures.

3.8 Resistance to oxygen-pressure aging. Valve disks shall show no cracking or increased hardness or tackiness after 72 hours of accelerated aging in oxygen, at a pressure of 2100 ± 100 Kilonewton per square meter and a temperature of 70° ± 1° C, when tested as specified in 4.2.4.5.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the disks of one size produced by one manufacturer from one batch (see 6.4) of raw materials and manufactured under essentially the same manufacturing conditions.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For destructive tests. Sheet stock produced in accordance with 3.1 shall be taken for the tests.

4.2.3 Examination.

4.2.3.1 For nondestructive tests. Sample disks shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.2.3.2 For destructive tests. The specimens shall be tested in accordance with 4.2.4.2. The acceptance number is zero.

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4.2.3.3 Classification of defects.

(a) Valve disks (see applicable dwg for size).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
*101	Leakage	4.2.4.3
102	Airflow resistance	4.2.4.4
	AQL 2.5 percent defective	
103	Flexibility (3.3)	
104	Thickness incorrect	
105	Center hole location incorrect	
106	Center hole diameter incorrect	
107	Disk diameter incorrect	
108	Tackiness	
109	Workmanship (3.7)	
Minor:	AQL 4.0 percent defective	
201	Color incorrect	
202	Finish incorrect	

*For prime procurement only.

4.2.4 Tests.

4.2.4.1 Preparation of test specimens. Specimens shall be prepared from the molded sheet stock (4.2.2.2) as specified in FED-STD-601. Specimens shall not be buffed or stretched before testing. Specimens shall be prepared for each test in the numbers indicated in table II. The test specimens selected for heat aging shall be aged in an oven at a temperature of 158° ± 1.8°F. for a continuous period of 168 ± 1 hour.

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Table II. Number of specimens for various tests

Test	Specimens	
	Unaged	Aged
Tensile strength	10	10
Ultimate elongation	10	10
Tension set	6	--

4.2.4.2 Physical properties tests. Tensile strength, ultimate elongation, tension set, shall be determined as specified in FED-STD-601.

4.2.4.3 Leakage. The valve disks shall be tested for leakage using Q211 flow type air leakage tester after they are assembled as follows:

Inlet valve disk (Dwg A5-3-95)-Assemble to inlet valve seat, 5-3-450, and test using DL136-14-170-5.
Inlet valve disk (Dwg A5-2-537)-Assemble to canister mounting, 5-2-535, and test using DL136-14-170-4.

4.2.4.4 Airflow resistance. The valve disks shall be tested for airflow resistance using Q212 inlet and outlet valve resistance indicator after they are assembled as follows:

Inlet valve disk (Dwg A5-3-95)-Assemble to inlet valve seat, 5-3-450, and test using DL136-44-1-5.
Inlet valve disk (Dwg A5-2-537)-Assemble to canister mounting, 5-2-535, and test using DL136-44-1-6.

4.2.4.5 Resistance to oxygen-pressure aging. Valve disks shall be tested in accordance with ASTM D572.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking, interplant shipment (see 6.5). The disks shall be packaged and packed to provide adequate protection from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation. Marking on shipping containers shall be in accordance with FED-STD-123 and shall include the name and address of the prime contractor and contract number.

5.2 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the repair parts packaging data sheet which is identified by its National Stock Number.

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6. NOTES

6.1 Intended use. These disks are intended for use in canisters and canister mountings.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color of disks.
(c) Applicable drawings.
(d) Applicability of leakage test (see 4.2.3.3).

6.3 Valve seats. Valve seats required for conducting airflow resistance and leakage tests will be furnished by the Government.

6.4 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.5 Interplant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

6.6 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 216. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army-EA

Preparing Activity:

Army-EA

Project No. 4240-A479

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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